

A PROJECT TITLE
ON
“Customer Churn Prediction Using Random Forest and XGBoost”
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

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DEPARTMENT OF Computer science and Information Technology

Name of Title: Customer Churn Prediction Using Random Forest and XGBoost

1. Abstract:

Customer churn is one of the major challenges faced by businesses that provide services to customers. It occurs when customers stop using a company's products or services and move to another provider. Losing customers can affect business growth and revenue, so it is important for organizations to identify customers who may leave and understand the reasons behind their decision. This project, "Customer Churn Prediction Using Random Forest and XGBoost," aims to use machine learning techniques to predict customer churn and provide useful insights into customer behavior.

The proposed system analyzes customer-related information such as tenure, contract type, monthly charges, payment method, services used, and other relevant details. The dataset is first cleaned and prepared through data preprocessing, followed by Exploratory Data Analysis (EDA) to understand customer patterns and identify factors that may influence churn. Based on the prepared data, two machine learning algorithms, Random Forest and XGBoost, are implemented to learn from previous customer behavior and predict the likelihood of customer churn.

The performance of both models is evaluated and compared using suitable machine learning evaluation measures to determine which algorithm provides better predictions for the given dataset. The project also focuses on understanding the important factors that contribute to customer churn, which can provide valuable information for businesses.

To make the project more practical and user-friendly, the trained machine learning model can be integrated into a web-based application. The application can allow users to provide customer details and view the predicted churn risk along with useful customer insights. This makes the project more than just a machine learning model by connecting data analysis, machine learning, and web development into a practical application.

Overall, the project demonstrates how machine learning can be applied to a real-world business problem. By identifying customers who are more likely to churn, the proposed system can help businesses take timely actions, improve customer satisfaction, strengthen customer relationships, and develop better customer retention strategies.

2. INTRODUCTION:

In the modern business world, customers play a major role in determining the growth and success of an organization. Companies invest considerable time and resources in attracting new customers and maintaining relationships with existing ones. However, customers do not always continue using the same products or services. They may change to another company because of pricing, service quality, customer experience, changing needs, or better alternatives available in the market. This situation, where a customer stops using a company's product or service, is commonly known as customer churn.

Customer churn is particularly important for businesses that depend on regular or long-term customers, such as telecommunication companies, banks, insurance providers, online platforms, and subscription-based services. A high rate of customer churn can affect a company's revenue, customer base, and long-term growth. At the same time, simply knowing that customers have already left does not help a business prevent the loss. Therefore, understanding customer behavior and identifying possible churn at an early stage has become an important business requirement.

With the rapid growth of digital services, organizations collect large amounts of information about their customers. This information may include their usage patterns, duration of association with the company, type of service, payment details, spending behavior, and other characteristics. Such data can provide valuable information about customer preferences and behavior. However, analyzing large datasets manually can be difficult and may not reveal hidden patterns effectively.

This is where Machine Learning becomes useful. Machine Learning enables computer systems to learn patterns from previously collected data and use those patterns to support predictions and decision-making. Instead of depending only on assumptions or manual analysis, organizations can use customer data to understand which characteristics are commonly associated with customers leaving a service.

Customer churn prediction is therefore an important real-world application of machine learning. It combines data analysis, customer behavior, and predictive modeling to help organizations make informed decisions. By identifying customers who may be at risk of leaving, businesses can focus their efforts on improving customer experience and developing suitable retention strategies.

This project explores customer churn prediction using Random Forest and XGBoost, two machine learning approaches that can be applied to customer data to study churn behavior. The project further aims to present the prediction process through a practical web-based application, connecting machine learning with a user-friendly real-world solution

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10/11, Linux, or macOS
- Programming Language: Python 3.x
- Development Environment: Jupyter Notebook / Google Colab / VS Code
- Data Processing: Pandas, NumPy
- Data Visualization: Matplotlib, Seaborn
- Machine Learning: Scikit-learn
- Machine Learning Model: Random Forest, XGBoost
- Dataset Format: CSV
- Web Browser: Google Chrome, Microsoft Edge, or any modern web browser


Signature of the Guide


Course Instructor